

Intermodulation spectroscopy and the nonlinear response of two-level systems in superconducting coplanar waveguide resonators

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Two-level system (TLS) loss is typically limiting the coherence of superconducting quantum circuits. The loss induced by TLS defects is nonlinear, resulting in quality factors with a strong dependence on the circulating microwave power. We observe frequency mixing due to this nonlinearity by applying a two-tone drive to a coplanar waveguide resonator and measuring the intermodulation products using a multifrequency lock-in technique. This intermodulation spectroscopy method provides an efficient approach to characterizing TLS loss in superconducting circuits. Using harmonic balance reconstruction, we recover the nonlinear parameters of the device-TLS interaction, which are in good agreement with the standard tunnelling model for TLSs.

I. INTRODUCTION

Superconducting quantum computing is facing a challenge in the relatively short coherent lifetimes of state-of-the-art qubits, currently in the sub-millisecond range [1–5]. A significant source of decoherence for these qubits is dielectric loss caused by parasitic two-level system (TLS) defects [6–8], whose microscopic origin is a topic of ongoing active research [9–11]. These TLSs can couple to qubits, providing a source of energy loss and parameter fluctuations, and therefore decoherence. In this article we introduce a method for the measurement and analysis of the inherent nonlinearity of these TLSs in superconducting resonators.

Superconducting resonators are essential components of quantum processors and a good testbed for characterizing dielectric loss. In circuit quantum electrodynamics resonators are treated as linear elements, typically modelled as lumped-element LC oscillators. The resonator's coupling to TLS defects gives rise to a characteristic dependence of the internal quality factor Q_i on circulating power, with a drop in Q_i at low powers where the TLS bath is not driven to saturation [12, 13]. This power-dependent response makes resonators ideal for probing TLSs in a qubit's environment.

In this work we explore how the power-dependent behaviour of Q_i translates to a low-power nonlinear response of the resonator. We measure this nonlinearity in a high-quality-factor coplanar waveguide (CPW) resonator, showing that it gives rise to frequency mixing, or intermodulation – a characteristic feature of nonlinear response which has been observed for TLS-induced [14] and other nonlinearities in superconducting circuits [15–17]. Intermodulation results in response at integer combina-

tions of the applied drive frequencies, $f_{\text{IMP}} = \sum_i k_i f_i$, where f_i are drive frequencies and k_i integers. The amplitude and phase of these intermodulation products (IMPs) can be measured with large signal-to-noise ratios (SNR) using lock-in techniques. Modelling the system as a driven harmonic oscillator with nonlinear damping due to parasitic TLSs, we reconstruct the nonlinear response of the device as a function of drive amplitude and frequency using harmonic balance analysis.

II. EXPERIMENTAL METHODS

We perform all measurements on a superconducting aluminium $\lambda/4$ coplanar waveguide resonator with a resonance frequency of 4.11 GHz, capacitively coupled to a transmission line in a notch configuration. The device has no engineered nonlinear elements. All microwave measurements presented here were carried out inside a dilution refrigerator at ~ 10 mK using the set-up outlined in Appendix B.

For the initial characterization of the resonator, we measure the S_{21} scattering parameter in transmission using a vector network analyzer and extract the relevant parameters such as Q_i and the resonance frequency f_r using the circle-fitting method [18], as detailed in Appendix D. We characterize Q_i for excitation powers ranging from the single photon level, relevant for quantum computing applications, up to 10^8 photons where we observe other, non-TLS related loss and nonlinear mechanisms.

The average number of photons $\langle n \rangle$ circulating in the resonator is estimated using the relation

$$\langle n \rangle = \frac{Z_0}{Z_r} \frac{Q_i^2}{|Q_c|} \frac{2P_{in}}{\hbar(2\pi f_r)^2}, \quad (1)$$

where Z_0 and Z_r are the characteristic impedances of the transmission line and the resonator, respectively. The

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